

Selina Liu

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OBJECTIVE

Full-stack web developer with hands-on experience designing and building scalable, accessible, and user-focused applications. Skilled in React, JavaScript/TypeScript, Node.js, Flask, SQL, and backend systems architecture. Passionate about clean, maintainable code and collaborating across the stack to solve complex problems. Open to remote or in-person roles in the Ann Arbor or Metro Detroit area.

EXPERIENCE

Web Developer

September 2024 – present

University of Washington Biology Department

Seattle, WA

- **Resolved 20+ production-level bugs** by refactoring legacy logic, introducing error tracking, and improving testing coverage, which **cut support requests by 100%** from stakeholders.
- **Credited as a co-author** in bioRxiv and under review for G3 for software contributions to a gene-editing research platform used by faculty and PhD scientists, improving research accessibility and enabling faster data dissemination.
- Designed and implemented a secure “Delete” function with access controls to support the compliant removal of **10,000+ genes**, significantly improving data management efficiency for researchers.
- Authored technical documentation on SQL/IP access configuration, cutting developer onboarding time by **over 40%** and improving maintainability of backend infrastructure.
- Refactored legacy React frontend by modularizing UI into reusable components, integrating updated packages, and implementing Redux for scalable state management and improved maintainability.

ECA Student Representative

October 2024 - Present

AIMS (Association of Information Management Students)

Seattle, WA

- Built a professional relationship with the Co-founder and Senior Product Manager of Quadrant Technologies and organized a high-impact networking event, successfully bringing together **70+ MSIM students** and **20+ Quadrant employees** for career development and industry insights in cloud computing.

Data Management Intern

May 2023 – August 2023

Procter & Gamble

Cincinnati, OH

- Implemented front-end development using **React.js** and connected it seamlessly to Node back-end through **SQL Server integration** and managing code base and documentation in **Azure DevOps**.
- Refactored and optimized legacy code, **reducing over 1,000 lines** across multiple files to **fewer than 200**, which improved maintainability, reduced technical debt, and accelerated future feature development.
- Implemented front-end architecture in React.js and collaborated on UX improvements, resulting in a **50% boost** in usability as measured by internal stakeholder feedback.
- Automated the product supply data hub table requesting process and streamlined payment and approval procedures by leveraging ServiceNow resulting in a **60% increase** in production rate.

PROJECTS

NASA Suits Challenge (Product Lead, AR Developer) | *Unity Game Engine, C#*

September 2022 – May 2024

- Engineered **3 core AR features** for NASA’s astronaut-assist interface by integrating Unity, WebSocket, and custom UI components, enabling real-time, bidirectional communication between the headset and mission control systems.
- Developed a dynamic in-helmet menu system, including an AI chatbot capable of delivering **30+ real-time, context-aware responses** to assist astronauts in accessing critical information during space missions.
- Built a **Unity 3D** front-end in **C#** that connected an AI chatbot to the MRTK HoloLens platform, enabling real-time, dynamic message exchange with synchronized speech and animation for immersive user feedback.
- Led **sprint planning, backlog grooming, and technical prioritization**, aligning cross-functional development efforts (UX, web, and AR) with evolving NASA mission requirements and test protocols.

EDUCATION

Master of Science in Information Management

Seattle, WA

University of Washington

September 2024 - August 2025

- GPA: 3.95
- Specialization: Business Intelligence
- Courses: Information Management, Strategic Leadership, Data Science Fundamentals, Portable Information Structures, Relational Database Management Systems, Business Intelligence Systems

Bachelor of Arts in Art & Design

Ann Arbor, MI

University of Michigan

September 2020 - May 2024

- Minors: Computer Science, UX Design
- Courses: Web Systems, Data Structure & Algorithm, Information Studies, Discrete Math, Interaction Design

SKILLS

Technical Languages: C++, C#, HTML, CSS, JavaScript, TypeScript, Python, SQL, R

Frameworks & Libraries: React.js, Next.js, Express.js, Flask, Node.js, MRTK, WebSocket

Developer Tools: VS Code, Visual Studio, R Studio, XCode, Git, NPM, Azure DevOps

Platforms & Technologies: Unity Game Engine, HoloLens, ServiceNow, Tableau, PowerBI, SQL Server

Practices: REST API Design, Agile Development, CI/CD, Version Control, Debugging, API Integration, Sprint Planning, Stakeholder Communication, Team Leadership, Prioritization, Cross-Functional Collaboration